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Upholding academic integrity: an exploratory study of AI-assisted detection of unauthorised machine translation use in student translations

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While machine translation (MT) offers potential benefits for language learning, concerns about academic integrity persist when MT outputs are incorporated by students into assessed work without prior permission or, where required, explicit disclosure. However, teachers' ability to detect such unauthorised MT use varies widely. This study explores whether providing raters with a consolidated analysis report from ProWritingAid (an AI-powered writing analytics tool) as a decision aid improves their detection accuracy across MT use conditions (non-MT-aided, Google Translate-aided, ChatGPT-aided). It also examines the linguistic indicators that support raters' detection judgements. A two-stage experiment was conducted at a Chinese university. Stage 1 involved 39 intermediate- to upper-intermediate-proficiency students of English as a foreign language (EFL), who were randomly assigned equally across the three MT use conditions and completed two translation tasks accordingly. In Stage 2, 78 EFL instructors served as raters and were randomly assigned in equal numbers to the unassisted rater group (S2UR) and AI-assisted rater group (S2AR). Before the detection tasks, the S2AR Group was provided with a ProWritingAid report and a brief orientation on how to interpret it. For each task, raters classified the translation sample from Stage 1 as MT-aided or non-MT-aided and recorded the linguistic indicators underpinning their judgement. The results show that AI assistance increased mean detection accuracy from ~52% to ~75%, and detection accuracy differed by MT use condition (highest for ChatGPT-aided samples, lowest for Google Translate-aided samples). Raters prioritised MT-strength-based cues (i.e., linguistic features indicative of proficiency beyond the cohort's expected level) over MT-error-based cues in their judgements. This exploratory study provides preliminary evidence that AI assistance can support rater-led detection of unauthorised MT use in translation assessment. Implications include combining AI flagging with human review, calibrating detection criteria for each MT system, and articulating clear policies on acceptable MT use in language education.

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Introduction

In recent years, artificial intelligence (AI) has become increasingly influential in education, shaping a broad spectrum of disciplines ranging from the social sciences to the humanities (Alfarsi et al. 2024; Zawacki-Richter et al. 2019). Foreign language education has also seen rapid uptake of AI-powered tools (Derakhshan and Ghiasvand 2024; Tsai et al. 2024). Among these tools, machine translation (MT) plays a significant role in supporting the development of both receptive skills, such as reading and listening, and productive skills, such as speaking, writing, and translation (Klimova et al. 2023; Lee 2023; Lee and Kang 2024). While MT offers substantial pedagogical benefits, it also raises concerns about academic integrity, which refers to a commitment to upholding honesty, fairness, trust, and responsibility in all academic work (De Maio and Dixon 2022). Institutions and teachers increasingly perceive that excessive reliance on MT may blur the boundary between legitimate assistance and academic misconduct, a trend often described as “high-tech plagiarism” (Derakhshan and Ghiasvand, 2024). This normalisation of MT use can lead students to treat MT as broadly acceptable (Murtisari et al. 2024; Vinall et al. 2024), even when its use does not comply with assessment requirements and thus constitutes unauthorised MT use. Such unauthorised use may bypass key cognitive processes that are central to language learning (e.g., idea generation, problem-solving) and may mask students’ authentic performance, undermining fairness and trust in assessment (Gonsalves 2024; Murtisari et al. 2024). These concerns motivate our focus on detecting unauthorised MT use in students’ translations.

To operationalise this focus, we first clarify how “unauthorised MT use” is understood. Acceptability of MT use is task-dependent and varies with the extent of reliance on MT outputs, the mode of assessment, and the language skill assessed (Hellmich and Vinall 2021; Maimone et al. 2025; Vinall et al. 2024). In general, acceptability is higher for word- or phrase-level look-ups, oral presentations, and receptive skills (e.g., reading) than for sentence- or paragraph-level incorporation of MT outputs, written assignments, and productive skills (e.g., writing). Building on prior research, we operationally define unauthorised MT use as the incorporation of MT outputs at the sentence level or above into written translation assignments without prior permission or, where required, explicit disclosure.

Detecting unauthorised MT use is an important way for teachers to manage students’ MT use, thereby harnessing its pedagogical benefits while maintaining academic integrity (Jolley and Maimone 2025). The detectability of MT use in student submissions, however, remains contested. Previous studies have revealed substantial variation in teachers’ ability to detect student MT use, with detection accuracy varying from very low (e.g., Stapleton and Leung 2019) to moderately high (e.g., Innes 2019). One likely contributor is the limited training and guidance available to teachers, as evidenced by a scarcity of research on the topic (Jolley and Maimone 2025). Compounding the challenge is the absence of consensus on linguistic indicators signalling MT use in student work: some studies focus on MT weaknesses (e.g., Innes 2019) while others identify MT strengths as potential indicators (e.g., Ducar and Schocket 2018). The wider availability of advanced MT systems such as ChatGPT has further complicated MT use detection, as different user prompts can yield translations of markedly different quality (Gao et al. 2024; Sahari et al. 2023). Given the high likelihood that students will continue to use MT in their learning (Stapleton and Leung 2019), and the substantial variability in teachers’ ability to detect such use (Maimone and Jolley 2023), it is essential for the academic community to explore tools that can effectively support teachers in identifying unauthorised MT use in student submissions.

Against this unsettled picture, and given the growing use of AI-powered tools in educational assessment (Fabiya 2024; Farrokhnia et al. 2024; Zawacki-Richter et al. 2019), the present study explores whether ProWritingAid, an AI-powered writing analytics tool with integrated textual analysis and plagiarism checking features (Pitukwong and Saraiwang 2024), can assist teachers in distinguishing between MT-aided and non-MT-aided student translations when provided as a decision aid. In this paper, MT-aided translations refer to student translations produced with assistance from an MT system (e.g., Google Translate or ChatGPT), whereas non-MT-aided translations refer to translations produced manually without MT assistance. Specifically, we compare detection outcomes across three MT use conditions (non-MT-aided, Google Translate-aided, ChatGPT-aided) and elicit the linguistic indicators that raters cite when making detection judgements. While previous literature has mainly focused on human-rater-only approaches to detecting MT use in translation assessment (e.g., Correa 2014; Ducar and Schocket 2018; Innes 2019), this exploratory study investigates whether AI-powered tools can assist teachers in detecting unauthorised MT use in student translations. The paper proceeds as follows: Section “Literature review” reviews related work and presents the analytical framework for MT use detection; Section “Methods” details the experimental design; Section “Results” presents the results; Section “Discussion” discusses underlying mechanisms and pedagogical implications; Section “Conclusion and future directions” summarises the main findings and outlines limitations and directions for future research.

Literature review

Within language education, covert MT use may compromise assessment validity and fairness (Gonsalves 2024; Tsai et al. 2024). Accordingly, detection of unauthorised MT use is positioned as an important measure for upholding academic integrity, particularly with respect to honesty, fairness, and trust in academic work (De Maio and Dixon 2022; Jolley and Maimone 2025). Key strands of research in this field include teachers’ ability to detect unauthorised MT use, linguistic indicators signalling student MT use, and an analytical approach to comparing MT-aided and non-MT-aided translations. Drawing on these strands, the present study synthesises prior work into an analytical framework for distinguishing MT-aided from non-MT-aided translations.

Teachers’ ability to detect unauthorised MT use. The empirical literature to date has focused mainly on teachers’ unassisted detection of student MT use (i.e. detection without external tool support), whereas tool-supported detection has only recently begun to attract research attention (Maimone et al. 2025). Across studies, reported findings indicate substantial variability in teachers’ ability to distinguish MT-aided work from independently produced student work, with accuracy rates ranging from low to moderately high (e.g., Maimone and Jolley 2023; O’Neill 2013; Stapleton and Leung 2019). For instance, Stapleton and Leung (2019) found that only two of the twelve teachers initially suspected the use of MT in student essays before they were informed that some essays were MT-aided. By contrast, in a translation exercise with English as a foreign language (EFL) learners, Innes (2019) reported a mean detection accuracy of 74.04% for identifying Google Translate-aided outputs. Other studies likewise report average detection accuracy rates around 70% across writing and translation tasks (e.g., Correa 2014; O’Neill 2013; Maimone and Jolley 2023). Conducted under unaided conditions,

these studies collectively indicate substantial heterogeneity in teachers' detection ability across tasks and study settings.

Several factors may affect teachers' detection accuracy. First, systematic rater training and calibration have often been limited, partly because the integration of MT into language education has only become widespread in recent years (Lee 2023). Consequently, many teachers may lack the knowledge and skills required to identify MT-aided work effectively, particularly when MT is used without authorisation. Second, institutional policies on acceptable and unauthorised MT use are inconsistently interpreted and enforced, leading to diverse classroom practices, ranging from word- or phrase-level look-ups to the incorporation of full-text MT outputs with or without human post-editing (Groves and Mundt 2021; Hellmich and Vinall 2023; Murtisari et al. 2024). Such variation in the degree of MT involvement in student work may, in turn, lower detection accuracy. Third, advances in neural MT have made MT outputs increasingly similar to human-produced texts (Popel et al. 2020; Yamada 2019), especially when students post-edit MT outputs, further complicating human judgement. Additionally, prompt sensitivity of large language models (LLMs) introduces within-system variability in output quality (Gao et al. 2024), thereby complicating unaided detection. Taken together, these constraints help explain why detection accuracy fluctuates across raters and tasks. Against this background, this study investigates whether AI assistance can enhance teachers' accuracy in detecting unauthorised MT use in student translations.

Linguistic indicators for detecting student MT use. Prior research has sought to identify linguistic indicators that signal MT use in student work, with a view to improving detection accuracy. Two lines of inquiry are prominent. One line focuses on MT weaknesses, treating MT errors as cues of MT use. Reported features include mistranslation, unidiomatic collocation, agreement or morphological errors, awkward yet grammatical sentences, referential inconsistency, and cohesion problems (Innes 2019; Steding 2009). Such cues align with evidence from comparative studies contrasting human-translated and machine-translated texts, which suggest that MT outputs, especially when only minimally post-edited, tend to exhibit recognisable patterns in morphology, lexical choice, syntax, collocation, and text-level cohesion (e.g., Frankenberg-Garcia 2022; Shin and Chon 2023). A second line focuses on MT strengths, drawing attention to linguistic features indicative of proficiency beyond what would be expected of the student cohort (i.e., MT-strength-based cues). Related studies have noted that unusually sophisticated lexis and syntax, dense use of connectives, enhanced cohesion and coherence, and unusually high surface accuracy can prompt suspicion of MT involvement (Ducar and Schocket 2018; Maimone and Jolley 2023; O'Neill 2013). This has the potential to shift raters' emphasis from MT-error-based cues to MT-strength-based cues when detecting MT use.

As MT technology evolves, scholars have debated whether increasing accuracy and sophistication change the relative usefulness of different indicators, potentially reducing the salience of overt MT errors and increasing the informativeness of MT-strength-based cues (e.g., Maimone and Jolley 2023). At the same time, students do not necessarily identify or correct typical MT errors accurately during post-editing (e.g., Zhang et al. 2024), so MT-error-based cues still persist. In short, technological change appears to shift the relative prominence of indicators rather than eliminate either line of evidence.

Building on this, emerging evidence suggests that differences between MT systems can also shape cue salience. Outputs from LLM-based systems (e.g., ChatGPT) have been reported to differ

from those of conventional MT systems (e.g., Google Translate) in lexical sophistication, sentence length, density of connectives, and overall quality (Gao et al. 2024; Rao et al. 2024; Sahari et al. 2023). Given the recency of LLMs in educational settings, relatively few studies have reported detection rates for LLM-aided translations under controlled conditions. To address this gap, the present study compares detection accuracy across MT use conditions and examines the linguistic indicators that raters associate with MT-aided translations.

Analytical framework for differentiating MT-aided from non-MT-aided translations. The analytical approach to translation quality assessment, which uses natural language processing (NLP) tools to quantify textual, syntactic and lexical indices in translations (Tian et al. 2025), can enable direct, reproducible comparisons between non-MT-aided and MT-aided translations. Previous research suggests that such indices can complement human judgement by quantifying differences in textual features that are less apparent in holistic scoring, and are increasingly used in fine-grained analyses of translations (e.g., Kwok et al. 2025). For example, Fu and Liu (2024) observed that post-edited MT outputs showed greater lexical diversity, longer mean sentence length, and a higher rate of passive constructions than human-translated texts. Accordingly, although such analytical indices do not, by themselves, constitute conclusive evidence of student MT use, they can serve as cues for distinguishing between non-MT-aided and MT-aided translations.

Within this approach, indicators are routinely quantified at three levels (i.e., textual, syntactic, and lexical), following established practice in translation quality assessment (e.g., Kwok et al. 2025; Liu and Sun 2024; Tian et al. 2025). Lexical measures capture sophistication and diversity (e.g., mean word length, lexical diversity, frequency-based familiarity); syntactic measures approximate structural complexity (e.g., mean sentence length, clause-combining ratios, nominal/phrasal density); and textual cohesion is estimated via connective density and referential overlap. This three-level profiling provides interpretable cues that can be aligned with rater expectations about students' language proficiency.

Building on this three-level assessment approach, an NLP-based analytical framework offers a principled basis for detecting student MT use. Tool-derived indices are treated as observable cues that can support rater judgements rather than as a stand-alone MT detector (Maimone et al. 2025). This framework is consistent with multi-level indicators documented in MT use detection research and has been reported to be effective in separating MT-aided from non-MT-aided texts in writing tasks (e.g., Chon and Chin 2020). Its potential in translation contexts is likewise evidenced by studies comparing linguistic differences between MT-aided and non-MT-aided translations. Meanwhile, among available NLP tools, ProWritingAid produces a standardised, consolidated report that maps onto these levels (e.g., grammar and spelling, readability, sentence length and variety, cohesion), thus enabling rapid and comparable profiling of texts (Banks and Lepki 2018; Ding and Zou 2024). Accordingly, the present study employs ProWritingAid as a decision aid to support rater-led judgements about potential MT involvement in student translations.

Research questions. A synthesis of Sections "Teachers' ability to detect unauthorised MT use", "Linguistic indicators for detecting student MT use" and "Analytical framework for differentiating MT-aided from non-MT-aided translations" highlights three observations that motivate the present study. First, in unassisted detection settings, teachers' ability to detect unauthorised MT use

varies markedly across raters and tasks. Second, earlier studies identify linguistic indicators at the lexical, syntactic, and textual levels that help distinguish non-MT-aided from MT-aided work and can be captured by NLP-based analytics reports in a transparent, reproducible form. Third, empirical evidence on the utility of AI-based analytics reports as decision aids for detecting unauthorised MT use in translation assessment remains limited, despite concerns about unauthorised MT use (e.g., Cotton et al. 2024; Mundt and Groves 2016) and emerging evidence of human-AI complementarity in language assessment (e.g., Fabiyi 2024; Shin and Lee 2024). Accordingly, the study compares rater judgements made with and without access to a consolidated ProWritingAid report, evaluates detection accuracy across MT use conditions, and records the linguistic indicators raters cite when forming their judgements. The research questions are formulated as follows:

- (1) To what extent does access to the consolidated ProWritingAid report improve raters' accuracy in distinguishing MT-aided from non-MT-aided translations?
- (2) How does detection accuracy vary across the three MT use conditions (non-MT-aided, Google Translate-aided, ChatGPT-aided) in the AI-assisted and unassisted rater groups?
- (3) Which linguistic indicators do raters cite as evidence of MT use?

Methods

The study was carried out during the first half of 2024, with data collected in two stages. In Stage 1, Chinese EFL learners produced translation samples after being randomly assigned in equal numbers to one of the three MT use conditions. In Stage 2, EFL instructors served as raters and classified the translation samples from Stage 1, making a binary classification (MT-aided vs non-MT-aided) and then selecting, from a structured list, the linguistic indicators that informed their judgements. The same set of translations was classified in parallel by raters in an AI-assisted condition (with access to a ProWritingAid report) and an unassisted condition (without access to the ProWritingAid report), allowing direct comparison across detection conditions. Ethics approval was granted by the research ethics committee of the first author's institution prior to data collection; all participants provided written informed consent.

Participants. Convenience sampling was employed in this study. In Stage 1, the participants were 39 EFL learners enrolled in an Integrated English Course at a Chinese university. The course, held twice a week for 90 min per session, aims to enhance students' language skills, including their translation skills. They had passed the College English Test Band 4 (CET-4), a national English proficiency test in China, with scores ranging from 425 to 630 (out of 710), which indicates intermediate to upper-intermediate English proficiency (Jin et al. 2022). None of them had received systematic training in MT use. The participants were randomly assigned to the three MT use conditions: non-MT-aided (S1NT Group, $N=13$), Google Translate-aided (S1GT Group, $N=13$), and ChatGPT-aided (S1CT Group, $N=13$). To verify whether participants from the three groups had comparable English proficiency, their CET-4 scores were used as a reference. The one-way ANOVA analysis revealed no significant differences in English proficiency among the three groups of students ($F(2, 36) = 0.113$, $p = 0.893$, $\eta^2 = 0.006$).

In Stage 2, 78 EFL instructors were recruited to classify the students' translations as either MT-aided or non-MT-aided. The raters were randomly assigned in equal numbers to the unassisted rater group (S2UR Group, $N=39$) and AI-assisted rater group

(S2AR Group, $N=39$). During the pre-task questionnaire, the raters self-rated their translation competence as proficient ($N=62$) or advanced ($N=16$). Regarding prior experience with MT, 73 reported previous use and 5 reported no experience. On whether using MT constitutes academic dishonesty, 15 respondents said it was not cheating, 59 said it depended on how MT was used, and 4 believed it was cheating regardless of how it was used. For perceived ability to detect student MT use, raters reported feeling not confident at all ($N=1$), not very confident ($N=2$), somewhat confident ($N=24$), confident ($N=50$), or very confident ($N=1$).

Description of the AI-powered tool used in the study. ProWritingAid is an AI-powered writing analytics tool that uses natural language processing to assess written texts and provide feedback (Banks and Lepki 2018; Pitukwong and Saraiwang 2024). In this study, it was used as a decision aid for raters, which surfaced interpretable linguistic features; human raters remained the final arbiters. Only raters in the AI-assisted detection condition (S2AR Group) received a consolidated ProWritingAid report for the six samples before making their detection judgements, while unassisted raters (S2UR Group) made their judgements without access to the report. Before the detection tasks, the S2AR Group received a brief orientation on how to interpret the ProWritingAid analytics report, including metric definitions. We deliberately did not prescribe metric cut-offs or algorithmic decision rules, as ProWritingAid has not been widely validated for MT detection. Raters were instructed to use the ProWritingAid report as supplementary cues alongside expectations for intermediate- to upper-intermediate-level EFL learners, rather than as determinative proof.

Among the report types available in ProWritingAid, the consolidated report supplied to the S2AR Group summarised the metrics in Table 1, with operational definitions adapted from the ProWritingAid eBook (Banks and Lepki 2018). Grammar and Spelling is an overall correctness percentage (0–100%) based on automated checks for grammar, spelling, and punctuation, with higher percentages indicating fewer issues and text closer to error-free. Readability Grade gives a grade-level estimate of reading difficulty using standard readability formulae (e.g., Flesch Reading Ease), where higher grades denote greater textual complexity. Transition Usage is the percentage of sentences that contain an explicit transition word or phrase (0–100%), with greater percentages reflecting more explicit cohesion. Average Sentence Length is the mean number of words per sentence, with a larger mean corresponding to longer sentences. Sentence Variety Score is the standard deviation of sentence length in words, with larger values reflecting greater variation in sentence length. As sentence variety involves changing the length, structure, and kinds of sentences to avoid monotony, it is often regarded as an important component of language proficiency (Charpentier-Jiménez 2020).

Table 1 shows a sample consolidated ProWritingAid report (pre-task reference), in which the six translation samples to be classified are profiled in a single view, enabling direct comparison of their textual properties. These are descriptive linguistic metrics, not detection measures. For example, in this sample, the MT-aided samples generally outperform the non-MT-aided samples in both translation tasks in terms of Grammar and Spelling, Readability Grade, and Transition Usage, whereas Average Sentence Length and Sentence Variety Score are broadly similar across conditions. Within the MT-aided conditions, the profile for Google Translate-aided sample is generally closer to the non-MT-aided profile than to the ChatGPT-aided profile.

Data collection procedure. The experiment was conducted in two stages (see Fig. 1 for an overview). Stage 1 involved 39 EFL

Table 1 Sample ProWritingAid report for the S2AR Group (pre-task reference).

Metrics	Task 1				Task 2			
	Non-MT-aided		Google Translate-aided		ChatGPT-aided		Non-MT-aided	
Grammar and spelling (%)	12.615	22.615	9.077	24.769	10.615	14.462	30.923	28.615
Readability grade	9.308	33.846	16.385	34.615	16.077	6.000	6.846	7.462
Transition usage (%)	30.077	16.154	3.692	3.231	3.231	31.538	36.154	45.462
Average sentence length	16.154	3.692	3.231	3.231	3.231	15.077	17.538	17.000
Sentence variety score	3.462	3.692	3.231	3.231	3.231	4.077	3.692	3.000

Use these metrics as supplementary cues alongside cohort-level expectations for intermediate-level EFL learners. No metric cut-offs are prescribed; avoid reliance on any single indicator and triangulate multiple cues. The analytics do not constitute definitive proof of MT use; any flag requires human judgement.

students. Participants first completed a pre-task questionnaire on demographics and language background, and then undertook two parallel Chinese-to-English translation tasks under one of the three MT use conditions: non-MT-aided, Google Translate-aided, or ChatGPT-aided. The source passages were selected from CET-4 translation tasks from previous administrations. As CET-4 is commonly used to benchmark intermediate proficiency, with a pass indicating intermediate-level attainment (Jin et al. 2022), texts of this type are pedagogically appropriate for intermediate- to upper-intermediate-level EFL learners. To ensure comparability, the two passages were selected to be broadly comparable in length, topic familiarity, and overall text complexity. Each task had a 30-min time limit, consistent with CET-4 timing. Students in the non-MT-aided condition translated manually, whereas those in the MT-aided conditions reviewed and post-edited the provided MT output, yielding 78 translation samples.

Stage 2 recruited 78 EFL instructors as raters to classify the 78 samples from Stage 1. Raters made a binary judgement for each sample (MT-aided vs non-MT-aided; no graded scale was used) and then selected perceived indicators of MT use from a structured list. Task materials comprised 13 versions, each containing six translation samples (two translation tasks × three MT use conditions), with each version balanced across tasks and conditions. Task versions were randomly assigned to raters, and samples were anonymised and presented in random order. Raters were blinded to the MT use condition of translation samples. Each task version was classified by three raters from the S2UR Group (unassisted) and three raters from the S2AR Group (AI-assisted); each rater completed one version (six samples), yielding 468 detection judgements in total (234 per group).

The structured list contained ten predefined indicators of MT use derived from prior studies (e.g., Chon and Chin 2020; Maimone and Jolley 2023) and tailored to the study context. There were five indicators based on MT strengths (i.e., absence of errors; excessive use of advanced vocabulary, of grammar structures, and of complex clauses; and unexpected levels of cohesion) and five indicators based on MT weaknesses (i.e., wrong choice of words, idiomatic and fixed expression errors, wrong grammar choice, syntactic errors, and contextual coherence errors). Raters could also select “Other” to enter additional linguistic features they considered as cues of MT use when making judgements. This design enabled a direct comparison between unassisted and AI-assisted detection judgements.

Data analysis. To address the first and second research questions regarding differences in detection accuracy rates, linear mixed-effects modelling was used at the decision level, with each rater-sample judgement coded as 0 = incorrect and 1 = correct. This approach was well-suited for the experimental setup due to its capability to statistically manage numerous variables that are often unmanageable in naturalistic settings (Liu and Zheng 2022). Fixed effects included rater group (S2UR vs S2AR), MT use condition, and their interaction. Random intercepts were specified for raters and translation samples to account for clustering. Results are presented as model-estimated accuracy rates for each rater group and MT use condition, together with differences in percentage points. All statistical analyses were conducted in IBM SPSS Statistics 26, with the significance level set at $p = 0.050$.

To address the third research question, which explores the linguistic indicators the raters associate with MT-aided translations, selections from the structured list were summarised by rater group. These selections were then analysed to identify patterns in the indicators of MT use cited by the raters. This analysis identified the key linguistic features on which raters relied when

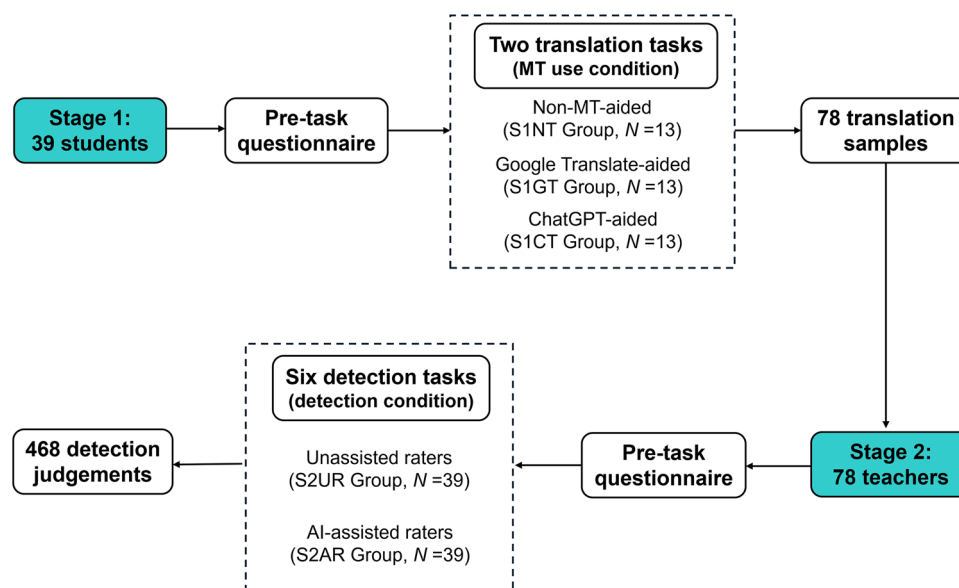


Fig. 1 Experiment procedure.

distinguishing between non-MT-aided and MT-aided translations.

Results

ProWritingAid and detection accuracy of MT use. The first research question asked whether the S2UR Group and the S2AR Group differed significantly in their detection accuracy. Results of the linear mixed-effects model showed a significant difference between the two rater groups, $F(1, 462) = 29.684$, $p < 0.001$, $f^2 = 0.064$. Parameter estimates further indicated lower detection accuracy for the S2UR Group than for the S2AR Group ($b = -0.167$, $SE = 0.073$, $t = -2.272$, $p < 0.050$). Descriptively, the mean detection accuracy was 52.1% for the S2UR Group and 75.2% for the S2AR Group. The results suggest that the raters identified students' MT use more accurately when supported by ProWritingAid analysis. See Table 2 for detection accuracy by rater group and MT use condition; confusion matrices and derived performance metrics (precision, recall/sensitivity, F1, specificity) are reported in Supplementary Appendix A.

MT use condition and detection accuracy of MT use. The second research question examined the influence of MT use condition on the accuracy of MT use detection by the S2UR Group and the S2AR Group. Results of the linear mixed-effects model showed a significant difference across MT use conditions, $F(2, 462) = 11.065$, $p < 0.001$, $f^2 = 0.048$. Parameter estimates further indicated lower detection accuracy for Google Translate-aided samples than for ChatGPT-aided samples ($b = -0.218$, $SE = 0.073$, $t = -2.971$, $p < 0.010$). To further clarify differences across the three MT use conditions, Bonferroni pairwise comparisons were conducted. These comparisons revealed a significant difference in detection accuracy between ChatGPT-aided and non-MT-aided translation samples ($p < 0.050$), as well as between ChatGPT-aided and Google Translate-aided translation samples ($p < 0.001$).

Descriptively (see Table 2), detection accuracy by MT use condition was 62.8% for non-MT-aided samples, 51.9% for Google Translate-aided samples, and 76.3% for ChatGPT-aided samples. The corresponding numbers of correct judgements were 98, 81, and 119, respectively. These differences should be interpreted in light of this study's context—English translations

of short Chinese source texts, intermediate- to upper-intermediate-level EFL learners, and heterogeneous post-editing behaviours, all of which may affect the detectability of unauthorised MT use. Patterns may differ with other tasks, cohorts, or post-editing practices.

Textual indicators associated with MT-aided translations. The third research question examined the linguistic indicators that the raters cited when classifying translations as MT-aided. Data comprised selections from a structured list completed immediately after each judgement. Raters who judged a sample to be MT-aided could select one or more indicators from ten predefined options, plus an "Other" option (total = 11). The ten options included five indicators associated with MT strengths and five with MT weaknesses (see "Data collection procedure" for details). The final option, labelled "Other," was open-ended, allowing the raters to specify additional indicators not covered by the predefined categories. Only selections linked to correct judgements were analysed. The selection details are shown in Fig. 2. For reporting clarity, the five MT-error-based subcategories are aggregated into a single "Indicators of MT errors" category. This aggregation does not alter the overall interpretation that both the S2UR and S2AR groups predominantly selected MT-strength-based cues as indicators of MT use; moreover, the individual error items were too infrequently chosen, with negligible between-group differences, to warrant informative item-level comparisons.

The data analysis shows that both the S2UR Group and the S2AR Group primarily relied on MT-strength-based cues rather than MT-error-based cues to identify student MT use. However, differences emerged in the overall number, pattern, and frequency of cue selections. The S2UR Group made 146 selections across correct judgements, compared with 284 in the S2AR Group. In addition, the S2UR Group exhibited a more concentrated selection pattern, relying disproportionately on a small subset of the five MT-strength-based cues. Namely, some cues were heavily favoured, whereas others were comparatively underused; for example, "Excessive use of advanced vocabulary" received 45 selections, and "Absence of errors" received only 15. By contrast, the S2AR Group showed a more even spread across the five MT-strength-based cues (roughly 40–62 selections each). Consistent with these distributional differences, the average

Table 2 Number of correct judgements by rater group and MT use condition.

Rating results	S2UR Group				S2AR Group			
	Non-MT-aided		MT-aided		Non-MT-aided		MT-aided	
	Google	ChatGPT-aided	Google	ChatGPT-aided	Google	ChatGPT-aided	Google	ChatGPT-aided
Correct	37 (47.4%)	53 (67.9%)	32 (41.0%)	53 (67.9%)	61 (78.2%)	66 (84.6%)	49 (62.8%)	66 (84.6%)
Incorrect	41 (52.6%)	25 (32.1%)	46 (59.0%)	25 (32.1%)	17 (21.8%)	12 (15.4%)	29 (37.2%)	12 (15.4%)
Total judgements	78	78	78	78	78	78	78	78

Correct judgements by rater group: S2UR Group 122 (52.1%; non-MT-aided = 37; MT-aided = 85); S2AR Group 176 (75.2%; non-MT-aided = 61; MT-aided = 115). Correct judgements by MT use condition: Non-MT-aided 98 (62.8%); Google Translate-aided 81 (51.9%); ChatGPT-aided 119 (76.3%). Percentages within each cell use 78 as the denominator.

number of cues cited per correct judgement was 1.718 (146/85) in the S2UR Group and 2.470 (284/115) in the S2AR Group. “Other” was rarely selected (S2UR: $N = 2$; S2AR: $N = 5$) and included cues outside the ten predefined categories (e.g., referential anomalies, over-literal rendering, sophistication inconsistencies). Given the small counts, no further analysis was undertaken.

This suggests that, in this sample, considering a broader cue set co-occurred with higher detection accuracy; no causal claim is implied. This pattern is consistent with the design of the AI-assisted detection condition: the S2AR Group had access to a consolidated linguistic feature report (e.g., grammar and spelling, transition usage, average sentence length), which likely broadened the range of cues considered before final judgement.

Discussion

The study investigated whether providing EFL instructors with a consolidated ProWritingAid report improved their accuracy in distinguishing MT-aided from non-MT-aided translations produced by intermediate- to upper-intermediate-level EFL students, and examined how MT use condition shaped detection accuracy as well as the linguistic indicators raters relied on during detection. The data analysis revealed several key findings: First, the S2AR Group achieved higher detection accuracy than the S2UR Group. Second, detection accuracy differed by MT use condition within this sample, highest for ChatGPT-aided translations and lowest for Google Translate-aided translations. Third, raters in both groups prioritised linguistic features perceived as too advanced for the students’ proficiency level over overt MT errors.

Role of ProWritingAid in enhancing detection accuracy. The results showed that access to the ProWritingAid report significantly improved raters’ detection accuracy of student MT use, with higher mean accuracy in the AI-assisted group (75.2%) than in the unassisted group (52.1%). While this accuracy level does not constitute a stand-alone basis for high-stakes decisions, the finding aligns with prior studies suggesting that AI-based analytics can support rater judgements (e.g., Fabiyi 2024; Farrokhnia et al. 2024). In this study, ProWritingAid was positioned as a decision aid rather than an MT detector. It surfaced a small, interpretable set of linguistic features that raters could combine with their expectations for intermediate- to upper-intermediate-level EFL learners, thereby likely reducing judgement noise under uncertainty. Taken together, these findings indicate that ProWritingAid can be usefully integrated into a rater-led workflow to support detection decisions. More broadly, the results are consistent with calls to integrate AI analytics judiciously to inform assessment while keeping final judgements human-led (Fabiyy 2024).

Despite these gains, two types of misclassification remained: non-MT-aided samples were classified as MT-aided (false positives), and MT-aided samples were classified as non-MT-aided (false negatives). This finding aligns with earlier studies indicating that MT output can exhibit human-like errors (e.g., Calvo-Ferrer 2023; Popel et al. 2020; Yamada 2019), thereby complicating the distinction between MT-aided and non-MT-aided texts. Because false positives and false negatives persisted even under the AI-assisted detection condition, any suspicion raised by AI-powered analytics should prompt evidence gathering and human review rather than an immediate attribution of unauthorised MT use. Keeping a few earlier translations for each student (a low-stakes personal portfolio) provides teachers with a personal reference point (Correa 2014), so that detection judgements are made with reference to the student’s own history

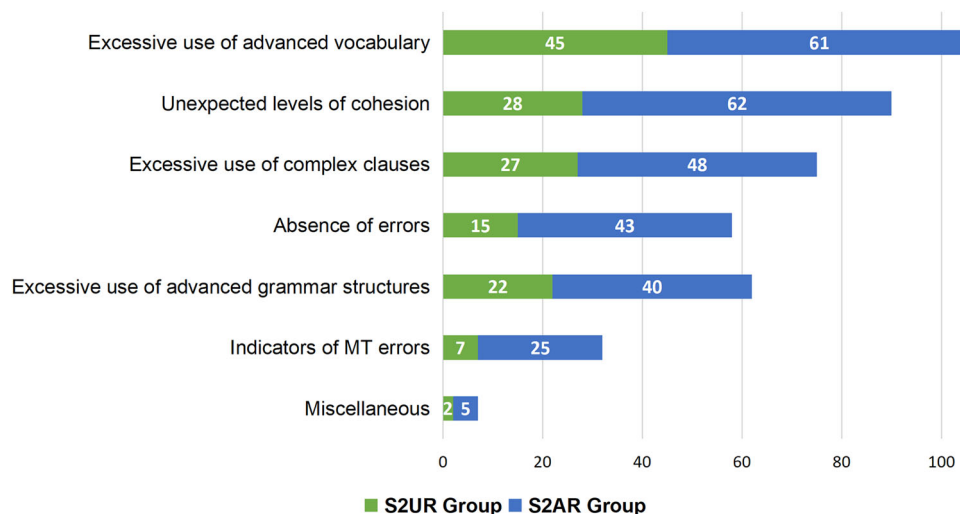


Fig. 2 Number of indicator selections by raters (by rater group).

rather than to a generic norm, with the aim of supporting more accurate, context-sensitive judgements.

Impact of MT use condition on detection accuracy. Detection accuracy varied across MT use conditions in this study's setting, being highest for ChatGPT-aided and lowest for Google Translate-aided translation samples, with non-MT-aided samples in between. This pattern is context-specific and should not be taken as evidence that translations produced with the aid of LLM-based MT outputs are easier to detect. A possible explanation is that ChatGPT-aided outputs displayed clusters of features that appeared unusually advanced for intermediate- to upper-intermediate-level EFL learners, consistent with findings that AI-powered MT systems can produce translations approaching professional quality (e.g., Popel et al. 2020). Such “advanced-for-level” cues were interpreted as signals of MT involvement by raters in this study. Meanwhile, Google Translate-aided texts, especially when lightly post-edited, may more closely resemble translations produced by intermediate-level learners (Yamada 2019), thereby reducing discriminability. Previous research likewise suggests that human translations and traditional MT outputs (e.g., Google Translate) can be difficult to distinguish (Popel et al. 2020). Beyond MT system differences, variation in the extent and style of post-editing across students (Hellmich and Vinall 2023; Murtisari et al. 2024), likely further blurred the boundary between non-MT-aided and MT-aided translations.

These context-specific differences underscore the need for pragmatic, context-sensitive guardrails and a human-in-the-loop detection approach, rather than treating technology as the sole determinant of detection accuracy. In light of the differential detection accuracy observed across MT use conditions in this study, instructors may calibrate detection criteria using recent exemplars from the MT systems students are most likely to use (Steding 2009) and consider how those MT systems shape the diagnostic salience of linguistic indicators. In addition, institutions can provide clear guidance and training on acceptable and ethical MT use (Hellmich and Vinall 2023; Shin and Chon 2023), so that MT use practices are more consistent and transparent, thereby reducing the confounding influence of heterogeneous post-editing practices on detection.

Perceived linguistic indicators of MT use. The results indicate that raters primarily cited MT-strength-based indicators (e.g., excessive use of advanced vocabulary, unexpectedly strong

cohesion, and an absence of errors) more often than MT-error-based indicators as evidence of student MT use. The finding echoes earlier studies where teachers rely on linguistic features judged to be beyond the cohort's level (e.g., Ducar and Schocket 2018; Maimone and Jolley 2023). While pedagogically intelligible, this “too-good-is-suspicious” heuristic may carry a fairness risk: legitimately strong work, uneven development across skills, or additional support (e.g., tutoring, and prior exposure to topic) can yield “advanced-for-level” translations without MT involvement. In contexts where norm-referenced detection criteria are used and baselines vary across classes, such indicators may inflate false positives, potentially undermining trust and disproportionately affecting high-achieving students. We therefore recommend calibrating judgements by reflecting on the context-specific salience of MT-strength-based and MT-error-based cues, so as to improve inter-rater consistency and mitigate false positives informed by “advanced-for-level” cues.

Another finding was that the S2AR Group drew on a broader set of indicators and cited more indicators per correct judgement than the S2UR Group. This aligns with the view that AI-powered tools can widen raters' attention to textual features (Shin and Lee 2024). In our setting, the consolidated ProWritingAid report highlighted differences in lexical, syntactic, and textual properties across the assigned samples, potentially reducing the need for raters to derive such cues from scratch. With these interpretable prompts available, AI-assisted raters attended to a wider cue set at the point of judgement. A broader, multi-cue basis may, in turn, help mitigate fairness risks associated with over-reliance on any single indicator.

Implications for language education. Grounded in the present findings, three practice-oriented implications emerge. First, AI-powered tools such as ProWritingAid should be framed as decision aids rather than arbiters. Given the persistence of false positives and negatives, any suspicion raised by AI-powered analytics should trigger a transparent, human-in-the-loop workflow rather than a summary verdict, including human review and proportionate process evidence (e.g., a brief translator's note on key choices). Programme policies can operationalise this by (i) defining acceptable versus unauthorised MT use with concrete, course-level examples, (ii) providing a simple disclosure pathway (e.g., short MT use statements on coversheets), and (iii) setting privacy-by-design practices, including data minimisation and access controls (Farrokhnia et al. 2024; Miao and Holmes 2023).

Clarifying the scope and limitations of AI-based analytics reports helps pre-empt the misconception that analytics constitute proof, aligning with calls for responsible, educative integration of AI in education (Shin and Lee 2024).

Second, it is important to regularly calibrate for MT-system-specific variability in detectability. Because detection accuracy differed across MT use conditions, teachers should attend to the specific MT systems students are likely to use (Steding 2009) and periodically update detection criteria as those systems evolve. Programmes can maintain a small, regularly updated collection of recent exemplars at the target proficiency that covers non-MT-aided, Google Translate-aided, and ChatGPT-aided outputs. Where feasible, they can collaborate with NLP researchers to develop ethically sourced, MT-system-specific corpora to support exemplar curation and periodic updates. Each exemplar should be annotated with indicators typically salient for its source. For instance, LLM-aided outputs may show more explicit cohesion and higher lexical sophistication, while lightly post-edited conventional MT outputs may more closely resemble the work of intermediate-level learners (Gao et al. 2024; Popel et al. 2020). These exemplars serve dual purposes: they support brief rater calibration to align expectations, and they enable low-stakes, in-class comparisons that build students' understanding of MT systems. This source-specific calibration recognises that post-editing and AI prompt design can blur boundaries and helps avoid technology-deterministic claims.

Finally, MT literacy training is advisable to sustain the reliability of MT use detection and to mitigate fairness risks arising from over-reliance on MT-strength-based cues. In this study, raters in both groups relied on MT-strength-based indicators. Although detection accuracy improved with AI assistance, misclassifications persisted. These findings warrant a shift from a purely reactive, detection-led approach to proactive MT literacy training. For teachers, such training should encourage reflective engagement with the set of indicators, using cohort-baseline comparisons and source-specific exemplars across MT use conditions to reduce over-reliance on any single family of indicators. For students, the training should define when and to what extent MT is authorised for each task (Ehrensberger-Dow et al. 2023; Luo et al. 2025) and normalise straightforward disclosure practices, thereby making detection less prone to misclassification. Beyond fairness considerations, an additional benefit for lower-proficiency learners is that it reduces ambiguity about acceptable assistance, discourages over-reliance that may lead to avoidance of language use, and helps to prevent hindrance to skill development (Murtisari et al. 2024).

Conclusion and future directions

In a controlled translation assessment task with intermediate- to upper-intermediate-proficiency EFL learners, providing raters with an AI-generated analytics report improved the detection accuracy of unauthorised MT use across three MT use conditions. Detection accuracy varied by condition, and raters in both rater groups relied more on MT-strength-based indicators than on MT-error-based indicators when forming judgements. These results indicate that interpretable analytics, when used as decision aids within rater-led workflows, can aid MT use detection without replacing human judgement.

Several limitations constrain these conclusions. First, the study was conducted within a single institution, which may limit the generalisability of the findings. Second, the evidence base was restricted to English translations of two short Chinese source texts (i.e., a single Chinese-English language pair), and detection was assessed using binary judgements; together, these design choices

may reduce representativeness and limit measurement sensitivity. Third, students' post-editing behaviours were not controlled and likely varied across submissions, increasing heterogeneity in the textual cues available for rater judgements. In addition, we relied on a single analytics tool, and the observed effects may therefore be tool-specific. Collectively, these constraints narrow the scope for generalisation and indicate that the observed improvements in detection accuracy, while meaningful, are unlikely to be sufficient on their own to support high-stakes decisions.

Future work should be comparative and context-sensitive. First, replication studies across language pairs, proficiency bands, and institutions would help assess the external validity and boundary conditions of the findings. Second, experimentally varying the extent of post-editing for outputs produced with different MT systems could clarify how MT system choice and post-editing extent shift the salience of detection indicators. Third, conducting comparative evaluations of alternative AI-powered analytics tools (e.g., Grammarly, DeepL Write) and LLMs (e.g., ChatGPT, DeepSeek) against ProWritingAid could identify which tools provide greater support to raters and under which conditions they are most beneficial. Finally, pre-task background checks indicated some variation in the raters' prior MT experience and confidence in detecting MT use. Although task versions were randomly assigned to minimise potential confounding from these factors, their potential effects on indicator selection merit further investigation, given that rater experience can influence consistency in evaluation (Neittaanmäki and Lamprianou 2024).

Despite these limitations, the study makes an exploratory contribution by providing granular evidence on how detection accuracy varies with both detection condition (AI-assisted vs unassisted) and MT use condition in translation assessment. Advancing the field will require the comparative, process-aware studies outlined above to develop robust, transferable guidance that aligns detection practices with pedagogical commitments and consistently upholds academic integrity. Sustained dialogue among educators, applied linguists, technology developers, and policymakers will also be valuable for shaping responsible practices for MT use in educational settings.

Data availability

The raw data are not publicly available due to participant confidentiality and ethics/consent commitments. De-identified data supporting the findings are available upon reasonable request from the first author, in accordance with participant confidentiality and privacy protections.

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Author contributions

X Zhou: Conceptualization; data collection; methodology; data analysis and interpretation; writing—original draft; writing—review & editing. X Wang: Supervision; writing—review & editing.

Competing interests

The authors declare no competing interests.

Ethical approval

Ethical approval was obtained from the Research Ethics Committee of Hunan University of Humanities, Science and Technology (Approval No.: 2024-004-SFS; approved on February 27, 2024) prior to data collection. All procedures were performed in accordance with the Declaration of Helsinki and relevant institutional and/or national guidelines and regulations for research involving human participants. The approval covered the full study protocol, including participant recruitment, questionnaire administration, and the collection and analysis of data.

Informed consent

Written informed consent was obtained from all participants on June 5, 2024, prior to data collection conducted from June 6 to June 12, 2024. Participants were informed of the study purpose and procedures, the intended use of their data, confidentiality measures, and their right to withdraw at any time without penalty. Consent covered participation and the use of de-identified data for analysis and publication of aggregated findings. No identifiable personal information is included in the manuscript or supplementary materials.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1057/s41599-026-06827-7>.

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